



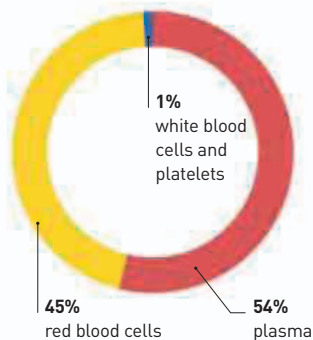
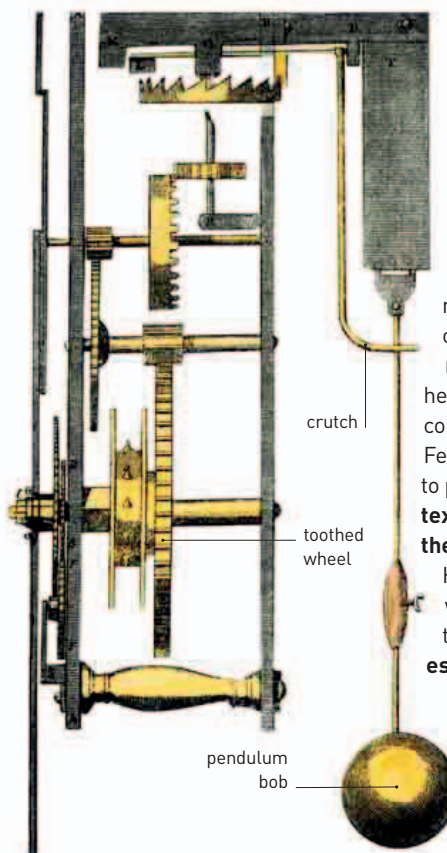
Dutch mathematician Christiaan Huygens was the first person to see Saturn's rings, and he suggested they were composed of solid particles.

IN 1655, JOHN WALLIS, AN ENGLISH MATHEMATICIAN, helped develop a way of finding the tangential lines to a curve—a fundamental aspect of the study of infinitesimal changes known as calculus. He devised a new mathematical **symbol to denote infinity** (a quantity larger than any number): ∞ . Four years later, Swiss mathematician

Johann Rahn would invent the **symbol for division**: $\div$. **Christiaan Huygens**, a Dutch mathematician and instrument-maker, invented new kinds of timepieces and telescopes. Early in 1655, he **discovered Titan**—Saturn's biggest moon—using a telescope he had made with his brother. By the end of 1656, he had noticed that Saturn's

crescents cast a shadow on the surface, suggesting that **these rings were made of solid material** not directly connected to the planet. In the same year, Huygens invented an **accurate pendulum clock**. Until the early 1600s, clocks could lose up to 15 minutes a day. Huygens' clock was a hundred times more accurate. By 1657, he was back to mathematics, collaborating with Pierre de Fermat and Blaise Pascal to publish the **first major textbook of probability theory**. The mechanism of Huygens' pendulum clock was further improved by the invention of the **anchor escapement** in 1657. Widely

Pendulum clock
Huygens' clock kept better time because the period of the pendulum remained the same, regardless of the amplitude of its swing.

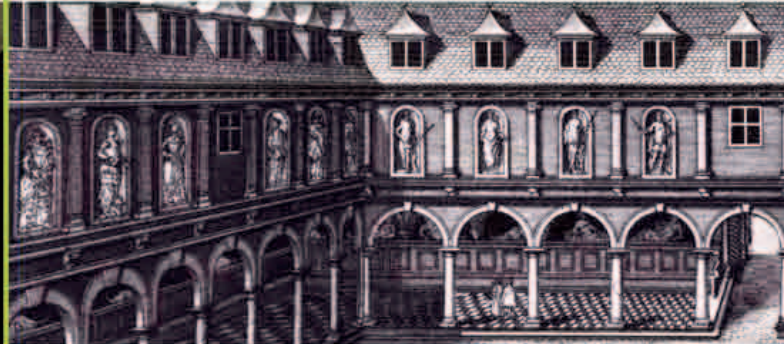


Components of blood by volume
The scarcity of white blood cells, along with inadequate microscopy, meant that 17th-century microscopists were able to record only red blood cells.

attributed to English inventor **Robert Hooke**, it allowed pendulum clocks to work with smaller swings and longer pendulums with heavier weights. Later, in 1658, Hooke devised the **balance spring for watches** as part of an improved escapement.

In 1657, a new scientific society was established in Florence, Italy. **Accademia del Cimento** (The Academy of Experiment) aimed to further enquiry by experimentation. Its prospectus became a popular laboratory manual in the 1700s.

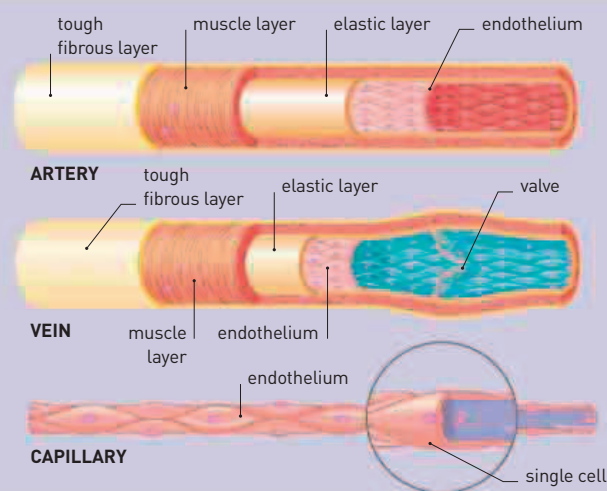
Jan Swammerdam, a Dutch biologist, spent much of his career studying anatomy and insects using a microscope. In 1658—before his university training—he was, purportedly, the **first person to observe red blood cells**.



Gresham College in London, England, was the original home of the Royal Society; the college was founded by financier Sir Thomas Gresham.

THE ROYAL SOCIETY, one of the oldest learned scientific societies, was founded in London in November 1660. The first meeting of 12 natural philosophers took place at Gresham College and included English architect **Christopher Wren** and **Robert Boyle**. The society met weekly to discuss “natural knowledge” and watch experiments; the first Curator of Experiments was Robert Hooke.

A year later, **Robert Boyle** published *The Sceptical Chymist*, which established his reputation as the **father of chemistry**. In it, he criticized the old alchemy and described a new scientific way of studying chemistry that advanced by experimentation. He replaced old ideas about nature's elements with the modern concept of an element as a pure substance that cannot be degraded into simpler forms.



TYPES OF BLOOD VESSELS

Thick-walled arteries have the most elastic fibers to help sustain the high pressure of blood from the heart. Thin-walled veins transport low-pressure blood and have valves to stop backflow as the blood returns to the heart. Between them are microscopic capillaries composed of endothelium (a single-celled layer) only, which facilitates the transfer of food and oxygen into tissues.

March 25, 1655
Christiaan Huygens discovers Titan, the largest moon of Saturn

December 1656
Huygens makes the first accurate pendulum clock

1657 Huygens publishes the first book on probability theory

1657 The Academy of Experiment is founded in Florence

1658 Hooke invents the balance spring for watches

1658 Jan Swammerdam possibly observes red blood cells

November 8, 1660
The Royal Society is founded in London

1660 Thomas Willis describes and names puerperal fever

1660 Robert Boyle publishes *New Experiments Physico-Mechanical*, including experiments with an air pump